



UNIVERSITY OF BARISHAL

Research Title:
**Integrated Spatial Assessment of Water Quality Index and Community Health
Risk in Salinity-Stressed Shyamnagar, Bangladesh.**

Submitted to:



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Title of the Research:

Integrated Spatial Assessment of Water Quality Index and Community Health Risk in Salinity-Stressed Shyamnagar, Bangladesh.

Background:

Coastal region of Bangladesh, particularly Shyamnagar Upazila in Satkhira District, is highly vulnerable to climate change and salinity intrusion caused by sea-level rise, reduced upstream freshwater flow, shrimp farming, and recurrent cyclones such as Aila and Amphan (Hossain et al., 2025). These events have damaged embankments and increased saline water intrusion into groundwater and surface water, creating a persistent freshwater crisis. Local communities rely on shallow and deep tube wells, pond sand filters, and surface ponds for water. However, increasing salinity and chemical contamination threaten drinking water, domestic use, and agriculture (Yeasmin et al., 2022). While the Water Quality Index (WQI) is commonly used to evaluate water suitability, most studies in coastal Bangladesh focus on hydro-chemical analysis and overlook spatial variability and community health impacts. Long-term exposure to saline and contaminated water is linked to health problems such as hypertension, skin disorders, and gastrointestinal illnesses (Ashrafuzzaman et al., 2023). Agricultural productivity is also affected by saline irrigation, increasing livelihood vulnerability. Despite these interconnected risks, few studies integrate spatial water quality assessment with health outcomes and agricultural stress.

Objectives

- a. To assess groundwater and surface water quality by calculating WQI and analyzing its spatial distribution using GIS techniques.
- b. To evaluate community health conditions and water-use practices and examine their relationship with water quality using statistical analysis.
- c. To assess the impact of water salinity and quality on domestic use and agricultural productivity.
- d. To develop an integrated geo-health risk zonation map identifying low, moderate, and high-risk areas.

Methods:

i. Data collection:

a. Field Survey and Water Sampling

A systematic field investigation and survey will be conducted to collect representative groundwater (tube well) and surface water (Gher) samples from the selected salinity-affected coastal region. Sampling locations will be selected using a stratified approach to ensure spatial coverage across different hydrogeological conditions, land-use types, and areas vulnerable to salinity intrusion. At each site, basic in situ parameters such as pH, Electrical Conductivity (EC), temperature, and salinity will be measured using calibrated portable instruments. Water samples will then be collected in pre-cleaned polyethylene bottles following standard water sampling protocols. For heavy metal analysis, samples will be preserved appropriately to maintain chemical stability. All samples will be properly labeled and transported under controlled conditions to the laboratory for further analysis. Relevant field information, including source type, well depth (if applicable), and surrounding environmental characteristics, will be recorded to support subsequent spatial and statistical analysis.

b. GPS-based Data Collection:

The geographic coordinates (latitude and longitude) of each sampling location will be recorded using a handheld Global Positioning System (GPS) device to ensure spatial accuracy. These geo-referenced data will be integrated into a Geographic Information System (GIS) platform for spatial analysis and mapping. The collected GPS data will enable the preparation of distribution maps of water quality parameters and the Water Quality Index (WQI), as well as the identification of salinity hotspots and high-risk zones. Accurate geo-referencing will ensure reliable spatial interpolation and strengthen the overall scientific validity of the study.

c. Interviews and Questionnaires:

A household survey was conducted to gather information on socio-demographic status, drinking water conditions, salinity exposure, health effects, and agricultural impacts. The survey included gender and occupation; drinking water source, availability, depth, distance, and color; perception of salinity changes and water treatment practices; occurrence of waterborne diseases and hypertension; and the effects of salinity on irrigation, crop or prawn production, and shifts from crop farming to aquaculture.

ii. Lab Experiment

Collected water samples will be analyzed to determine key physicochemical parameters, including pH, Electrical Conductivity (EC), Total Dissolved Solids (TDS), salinity, major cations (Na^+ , K^+ , Ca^{2+} , Mg^{2+}), major anions (Cl^- , HCO_3^- , SO_4^{2-}), and iron (Fe). In addition, heavy metals such as arsenic (As), lead (Pb), and chromium (Cr) will be tested due to their significant public health implications.

Standard laboratory procedures will be followed, with pH and EC measured using calibrated meters (Hach Multimeter) and chemical constituents analyzed through appropriate instrumental methods (e.g., AAS or ICP-MS). Quality control measures will be maintained throughout the analysis. Results will be compared with national and international drinking water standards to assess suitability and to support Water Quality Index (WQI) calculation and health risk assessment.

iii. Spatial and Statistical Analysis

a. Water Quality Index (WQI) Computation

Water quality assessment was conducted using the Water Quality Index (WQI) as an integrative measure to synthesize multiple physicochemical parameters into a single, interpretable metric. Groundwater and surface water samples were analyzed for pH, electrical conductivity (EC), total dissolved solids (TDS), salinity, dissolved oxygen (DO), major cations (Ca^{2+} , Mg^{2+} , Na^+ , K^+), and anions (Cl^- , SO_4^{2-} , NO_3^-). Each parameter was assigned a weight (W_i) reflecting its relative importance to human health and agricultural suitability, following established guidelines (Kothari et al., 2021) (Standard Methods for the, 1998). The quality rating (Q_i) for each parameter was computed as:

$$Q_i = \frac{C_i}{S_i} \times 100$$

where C_i is the measured concentration of the i^{th} parameter and S_i is the corresponding standard value from WHO or national guidelines. The sub-index (SI_i) was then calculated:

$$SI_i = W_i \times Q_i$$

The overall WQI was determined as the summation of all sub-indices:

$$WQI = \sum SI_i$$

WQI values were classified into categories—excellent, good, poor, very poor, and unsuitable—providing a robust framework for evaluating water quality status spatially and temporally.

b. GIS-based Spatial Mapping

Geospatial analysis was employed to visualize the spatial heterogeneity of water quality parameters and WQI values. Sampling locations were georeferenced using GPS, and attribute data were integrated into a GIS platform (ArcGIS/QGIS). Continuous surface maps were generated using spatial interpolation techniques, specifically Inverse Distance Weighting (IDW) and Ordinary Kriging, to predict parameter values in unsampled locations. IDW was expressed as (GISGeography, 2016) :

$$Z(x_0) = \frac{\sum_{i=1}^n \frac{Z(x_i)}{d_i^p}}{\sum_{i=1}^n \frac{1}{d_i^p}}$$

where $Z(x_0)$ is the estimated value at location x_0 , $Z(x_i)$ is the observed value at location i , d_i is the distance between x_0 and x_i , and p is the distance decay parameter. Spatial interpolation facilitated the identification of contamination hotspots, geographic patterns, and priority areas for intervention, enabling evidence-based water resource management.

c. Health Risk Modeling

To quantify potential health risks, survey-based epidemiological data were integrated with water quality indices. Household surveys captured water-use practices and incidence of water-related diseases (e.g., diarrhea, skin infections, respiratory issues). Health risk indices were modeled using regression-based approaches, where the risk level (R) was expressed as a function of water quality parameters:

$$R = \beta_0 + \sum \beta_i X_i + \epsilon$$

Here, X_i represents each water quality parameter or WQI, β_i are regression coefficients, β_0 is the intercept, and ϵ is the error term. Risk categorization into low, moderate, and high classes allowed identification of vulnerable populations and spatial clustering of high-risk zones, providing critical insight for targeted public health interventions.

d. Correlation with Agriculture and Water Quality and Risk Zoning Map

The interplay between water quality and agricultural productivity was analyzed to assess environmental and agronomic vulnerability. Data on irrigation practices, cropping intensity, soil salinity, and yield patterns were statistically correlated with WQI and individual parameters using Pearson or Spearman correlation coefficients (Statistics How To, 2026)

$$r = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum (X_i - \bar{X})^2 \sum (Y_i - \bar{Y})^2}}$$

Spatial integration of water quality, health risk, and agricultural sensitivity layers in GIS facilitated the generation of a composite **risk zoning map**. Overlay analysis classified the study area into distinct zones of low, moderate, and high risk, providing a comprehensive decision-support tool for integrated water resource management, agricultural planning, and climate-resilient public health strategies

Scope

The project aims to provide a comprehensive assessment of groundwater (tube well) and surface water (gher) quality in the study area to determine their suitability for drinking and domestic use. It will generate scientific evidence on physicochemical parameters and heavy metal concentrations, identifying potential environmental and public health risks. The study will be implemented through systematic sample collection, laboratory analysis, statistical evaluation, and comparison with national standards. Importantly, the findings will primarily support national development priorities by informing safe and equitable access to drinking water and addressing water-related health risks. While the results may contribute to global knowledge, the focus is on providing policy-relevant recommendations for sustainable water resource management and risk mitigation at the local and national levels.

Expected Results

- Significant spatial variation in groundwater and surface water quality.
- Several sites classified as poor to unsuitable for drinking (WQI-based).
- High salinity and TDS in coastal and intensively cultivated areas.
- Identification of contamination hotspots through GIS mapping.
- Significant links between poor water quality and water-related diseases.
- Negative impacts of salinity on agricultural productivity.
- Development of an integrated risk zoning map to support sustainable management aligned with United Nations SDG 6 and United Nations SDG 3.

Timeframe

The following Gantt chart shows the research activities and plans to be completed within the time frame.

Tasks	May June 2026	July 2026	August October 2026	November December 2026	January February 2026	March 2026	April June 202
Reviewing the literature, initial works and designing research.							
Developing questionnaires, pre-testing, and finalization							
Collecting data through different techniques and Giving entry of data, processing, and analysis							
Laboratory analysis of physicochemical parameters and heavy metals, with data entry and quality verification.							
Preparing draft report and seminar presentation							
Result							
Final report preparation, editing, and submission (including printing and binding).							

Financial Budget

Phase	Description of Work	Anticipated Costs (₳)
Phase One	Field survey, transportation, sample collection materials	12,000
Phase Two	Laboratory analysis (physicochemical parameters and selected heavy metals)	22,000
Phase Three	Data analysis, printing, binding, miscellaneous	2,000
	Honorarium of the Researcher	24,000
	Total	₳ 60,000

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